

Dockerized 3 -tier App Demo

docker compose up --build -d

```
sehtab@sehtab-Q550LF: ~/Dockerized_3-tier
sehtab@sehtab-Q550LF:~/Dockerized_3-tier$ nano docker-compose.yml
sehtab@sehtab-Q550LF:~/Dockerized_3-tier$ docker compose up --build -d
WARN[0000] /home/sehtab/Dockerized_3-tier/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 1.5s (21/21) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 1.02kB 0.0s
=> [backend internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 174B 0.0s
=> [frontend internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 163B 0.0s
=> [backend internal] load metadata for docker.io/library/node:20-alpine 1.1s
=> [frontend internal] load metadata for docker.io/library/nginx:alpine 1.1s
=> [frontend internal] load .dockerignore 0.0s
=> => transferring context: 78B 0.0s
=> [backend internal] load .dockerignore 0.0s
=> => transferring context: 82B 0.0s
=> [frontend 1/3] FROM docker.io/library/nginx:alpine@sha256:62ff2089abf5a9ed33bd232895bef5e22f7bb4b2006 0.0s
=> => resolve docker.io/library/nginx:alpine@sha256:62ff2089abf5a9ed33bd232895bef5e22f7bb4b200675cec49a5 0.0s
=> [frontend internal] load build context 0.0s
=> => transferring context: 62B 0.0s
=> [backend 1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a046 0.0s
=> => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb 0.0s
=> [backend internal] load build context 0.0s
=> => transferring context: 125B 0.0s
=> CACHED [frontend 2/3] COPY nginx.conf /etc/nginx/conf.d/default.conf 0.0s
=> CACHED [frontend 3/3] COPY index.html /usr/share/nginx/html/index.html 0.0s
=> [frontend] exporting to image 0.1s
=> => exporting layers 0.0s
=> => exporting manifest sha256:a1c4f0d3ba338bd1daf53fe2c92f99e8b2a9a238d03a72ce038e565aa4e1b79e 0.0s
=> => exporting config sha256:78e98081b1a25a518c460cf3d9dc16301ee31bce7aef9723d4da9175a7718dd5 0.0s
=> => exporting attestation manifest sha256:4855bb0dd02279d9e28e0f17651c0def2bbb8164cd76b862594c73077910 0.0s
=> => exporting manifest list sha256:42f91bd04aab4091ae4677ce9c639823c1597868eaa3a18ef9e420c48377ce12 0.0s
=> => naming to docker.io/library/dockerized_3-tier-frontend:latest 0.0s
=> => unpacking to docker.io/library/dockerized_3-tier-frontend:latest 0.0s
```

```
sehtab@sehtab-Q550LF: ~/Dockerized_3-tier
=> [backend 1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a046 0.0s
=> => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb 0.0s
=> [backend internal] load build context 0.0s
=> => transferring context: 125B 0.0s
=> CACHED [frontend 2/3] COPY nginx.conf /etc/nginx/conf.d/default.conf 0.0s
=> CACHED [frontend 3/3] COPY index.html /usr/share/nginx/html/index.html 0.0s
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=> => exporting attestation manifest sha256:4855bb0dd02279d9e28e0f17651c0def2bbb8164cd76b862594c73077910 0.0s
=> => exporting manifest list sha256:42f91bd04aab4091ae4677ce9c639823c1597868eaa3a18ef9e420c48377ce12 0.0s
=> => naming to docker.io/library/dockerized_3-tier-frontend:latest 0.0s
=> => unpacking to docker.io/library/dockerized_3-tier-frontend:latest 0.0s
=> CACHED [backend 2/5] WORKDIR /app 0.0s
=> CACHED [backend 3/5] COPY package*.json ./ 0.0s
=> CACHED [backend 4/5] RUN npm install --only=production 0.0s
=> CACHED [backend 5/5] COPY . . 0.0s
=> [backend] exporting to image 0.1s
=> => exporting layers 0.0s
=> => exporting manifest sha256:74342c1a0ac46a3f47e446a9652539a53b7071ea090c07b195aef1379a3a8240 0.0s
=> => exporting config sha256:695dcec3b945edeffef6238176f1d42246d5b4c9b39101ac5e397a6a1049d95f 0.0s
=> => exporting attestation manifest sha256:90982b191b2eaf6618ab8295935e7538d17c4c2a22151613709c994fc229 0.0s
=> => exporting manifest list sha256:4e51b69d59567aa0d666c36238a3057ed4268e3cfd6f0a2de9f1f8fe213458e8 0.0s
=> => naming to docker.io/library/dockerized_3-tier-backend:latest 0.0s
=> => unpacking to docker.io/library/dockerized_3-tier-backend:latest 0.0s
=> [backend] resolving provenance for metadata file 0.0s
=> [frontend] resolving provenance for metadata file 0.0s
[+] up 5/5
✔ Image dockerized_3-tier-backend Built 1.6s
✔ Image dockerized_3-tier-frontend Built 1.6s
✔ Container app_db Running 0.0s
✔ Container app_backend Running 0.0s
✔ Container app_frontend Started 0.6s
sehtab@sehtab-Q550LF:~/Dockerized_3-tier$
```

docker compose ps

```
sehtab@sehtab-Q550LF:~/Dockerized_3-tier$ docker compose ps
WARN[0000] /home/sehtab/Dockerized_3-tier/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME                IMAGE                                     COMMAND
SERVICE            CREATED          STATUS          PORTS          COMMAND
app_backend          sha256:a41420e7ef47d605929a388edc07760a03f9d0003d8d6de675c1951c39c8e310  "docker-entrypoint.s..."
backend             22 minutes ago  Up 22 minutes  5000/tcp
app_db               postgres:16-alpine                      "docker-entrypoint.s..."
db                  22 minutes ago  Up 22 minutes  5432/tcp
app_frontend         dockerized_3-tier-frontend              "/docker-entrypoint...."
frontend            6 minutes ago   Up 6 minutes   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
sehtab@sehtab-Q550LF:~/Dockerized_3-tier$
```

<http://localhost:8081/>

Three-Tier App Status

Test Backend Connection

```
{
  "status": "ok",
  "db_time": "2026-09-19T20:47:06.192Z"
}
```

```
docker exec -it app_db psql -U admin -d myapp -c "
CREATE TABLE IF NOT EXISTS users (
  id SERIAL PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
INSERT INTO users (name, email) VALUES ('Alice Smith', 'alice@example.com');
"

CREATE TABLE
INSERT 0 1
```

```
sehtab@sehtab-Q550LF:~/Dockerized_3-tier$ docker exec -it app_db psql -U admin -d myapp -c "
CREATE TABLE IF NOT EXISTS users (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
INSERT INTO users (name, email) VALUES ('Alice Smith', 'alice@example.com');
"
sehtab@sehtab-Q550LF:~/Dockerized_3-tier$ docker exec -it app_db psql -U admin -d myapp -c "SELECT * FROM users;"
 id |   name   |   email   |   created_at
-----+-----+-----+-----
  1 | Alice Smith | alice@example.com | 2026-09-19 21:20:07.25187
(1 row)

sehtab@sehtab-Q550LF:~/Dockerized_3-tier$
```

Essential Docker Command:

Docker Operations & Commands	
Action	Command
View running containers	<code>docker compose ps</code>
View combined logs	<code>docker compose logs -f</code>
View logs for a specific service	<code>docker compose logs -f backend</code>
Stop all containers	<code>docker compose stop</code>
Stop and remove containers & networks	<code>docker compose down</code>
Wipe database & start completely fresh	<code>docker compose down -v</code>